

Diana Opiyo
Data Scientist

(616)-445-0881 | kopiyodiana@gmail.com | Allendale, MI 49401

<https://www.linkedin.com/in/diana-opiyo-680b98309/>

<https://dianaopiyo.vercel.app/>

PROFESSIONAL SUMMARY

Results-driven statistics, machine learning, and data science professional with over 7 years of experience in academic teaching, research, and applied analytics. Proven ability to build machine learning algorithms, analyze data in statistical software, and support evidence-based projects in healthcare settings. Skilled in Python, SPSS, R, SQL, and survey design, with a strong background in statistical modeling and data-driven decision-making. Passionate about data literacy and applying analytics to solve real-world problems.

EDUCATION

Master's of Science, Data Science and Analytics, Grand Valley State University, Allendale, MI, January 2028

Master's of Science, Applied Statistics, Technical University of Mombasa, Mombasa, Kenya, November 2026

Bachelor's of Science, Mathematics and Computer Science (Statistics Option), Technical University of Mombasa, Mombasa, Kenya, November 2019.

WORK EXPERIENCE

Research Assistant, Grand Valley State University, Department of Information Sciences and Technologies, May-August 2026

- Support data analysis and reporting for research projects in the Department of Information Sciences and Technologies.
- Prepare clear summaries, tables, and visualizations to communicate research findings.
- Contribute to research workflows through careful data cleaning, documentation, and interpretation.
- Collaborate with Dr. Ali Raza and department stakeholders to strengthen analysis quality and reporting clarity.

Instructor, Technical University of Mombasa, Kenya, December 2019–Present

- Delivered undergraduate lectures in statistics and mathematics and computer science courses.
- Conducted practical lessons using statistical software such as Python, SPSS, R, Excel, and STATA
- Supervised student research projects involving data collection, analysis, and reporting
- Conducted data analysis using PowerBI, Tableau, and SPSS.
- Guided students on how to use data collection tools and how to accurately collect data.
- Mentored students academically and professionally, offering support on research and career guidance.
- Participated in curriculum development within the department.
- Conducted mentorship programs to high school students within Mombasa region to promote STEM education and to encourage them to take STEM subjects seriously.

Data Analyst, Samaria Medical Center, Kenya, May 2019–December 2021

- Collected, managed, and analyzed clinical and operational data to support hospital decision-making
- Developed dashboards and summary reports using tools like Excel, SPSS, and R
- Collaborated with medical and administrative teams to track patient outcomes, staff performance, and resource use
- Ensured data integrity and confidentiality in handling patient and facility information
- Ensured security and backup of patient records in compliance with health data standards
- Helped with hospital report preparations that were presented to the ministry of health.

Research Assistant, Technical University of Mombasa Enterprises Limited, Kenya, January 2020–December 2020

- Part of the research team that successfully developed and won a Ministry of Education–funded proposal worth Ksh 10 million (~\$100,000) in government funding for a national tracer study on TVET graduate employability under the State Department for University Education.
- Conducted advanced data analysis using R and Python, managing large datasets and generating insights to inform curriculum and policy reforms.
- Contributed to the design and implementation of a national graduate employability database, now used by the Ministry of Education for ongoing policy evaluation.
- Collaborated with statisticians, lecturers, and ministry officials to ensure data accuracy and compliance with reporting standards.

Data Analyst, Technical University of Mombasa Enterprises Limited, Kenya, March 2018–June 2020

- Designed, collected and cleaned graduate survey data.
- Analysed and reported on graduate employment rates and trends.
- Ensured data accuracy and compliance with reporting standards.
- Provided data insights to inform programme and curriculum review.
- Conducted data analysis and visualization using R, PowerBI, Python, Tableau.

PUBLICATIONS

Opiyo, D. & Sawesi, S. (2026). Machine Learning-Based Prediction of Heart Disease Using a Merged UCI Dataset and Web Deployment. Accepted at the 5th International Conference on Innovations in Computing Research (ICR'26), Berlin, Germany.

Opiyo, D. & Sawesi, S. (2026). Predicting Perceived Medication Management Difficulty as an Early Indicator of Non-Adherence Risk: A Machine Learning Approach for Pharmacist Intervention.

Thesis: Deep Learning Model on Suicidal Ideation Detection Using Social Media Data

PROJECTS

MindGuard AI: NLP-based deep learning tool for early detection of distress in social media text.

Heart Disease Predictor: Built, trained, and evaluated multiple ML algorithms. Deployed best-performing model as an interactive Python web application.

PharmAssist: Medication management difficulty prediction.

Tracer Study on TVET Graduate Employability database: Contributed to a government-funded national graduate employability, analyzed large datasets and helped develop a graduate tracking database used by the Ministry of Education.

AWARDS & HONORS

- **Honourable Mention, Predict 5 Contest (Graduate Student Category):** American Statistical Association (ASA), 2026
- **Best Overall Project, Innovation Day:** College of Computing, Grand Valley State University, 2026, *Machine Learning-Based Prediction of Heart Disease*
- **Emerging Ideas Track Runner-Up, Grand Rapids DeepTech Pitch Competition:** 2026, *MindGuard AI; \$600 Applied Computing Institute Development Support funding.*
- **Certificate of Recognition, Winter 2026 Graduate Showcase:** Graduate School, Grand Valley State University, 2026
- **International Student Experience Scholarship (Academic Excellence):** Grand Valley State University, Winter 2026
- **International Student Experience Scholarship (Campus Community):** Grand Valley State University, Winter 2026

SKILLS

SPSS	Tableau	Access	Ethical Judgment Turn-taking
Data analysis	Pivot tables	Empathy	Time management
JASP	pandas	Collaborative	Problem solving
MS Excel	ggplot	Conflict resolution	Willingness to learn
Computer programming	Self-motivated	Adaptability	Accountability
Reliable	STATA	Organized	Good communication
R	Power BI	Critical thinking	PyTorch
SQL	ANCOVA	Confidential	TensorFlow
JAMOVI	scikit learn	Active listening	Huggingface
ANOVA	Time series analysis	Detail oriented	
Exploratory data analysis	Problem solving	PySpark	
Negotiation skills	Excel	Huggingface	
Python	Data automation	Mentoring and coaching	

COURSES TAKEN

Machine Learning: Developed and evaluated predictive models using supervised and unsupervised learning techniques for classification, regression, and pattern recognition

Data Engineering: Design and maintenance of data pipelines for collecting, transforming, and storing structured and unstructured data to support analytics and modeling. Applied SQL, MongoDB and NoSQL in real world scenarios, especially with healthcare data

Bayesian Analysis: Application of probabilistic modeling and Bayesian inference to quantify uncertainty, update beliefs with new data, and support data-driven decision making

Data Mining: Extraction of meaningful patterns, trends, and relationships from large datasets using statistical and computational methods.

Design and Analysis of Experiment: Planning and analysis of controlled experiments to assess causal effects, optimize processes, and ensure statistically valid conclusions